Chapter 70

Editing Parameters in the Inspector

The Inspector is where you adjust the parameters of each node to do what needs to be done. This chapter covers the various node parameters and methods for working with the available controls.

Contents

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Overview of the Inspector

While the creation and connection of nodes in the Node Editor determines the tools and order of operations that make up a composition, the Inspector is where you adjust the various parameters inside each node to do what needs to be done.

Inspector displays the Brightness Contrast controls

This chapter covers methods for opening node parameters in the Inspector to edit them in different ways according to the type of available controls.

To display the Inspector:

Click the Inspector button on the UI toolbar.

The Tools and Modifiers Panels

The Inspector is divided into two overall panels.

- The Tools panel is where the parameters of selected nodes appear so you can edit them.
- The Modifiers panel is where you edit optional extensions to the tool's standard toolset as well as automated expressions that you can attach to individual parameters to create animated effects. Additionally, certain nodes such as the Paint node generate data such as Strokes, which are saved in the Modifiers panel.

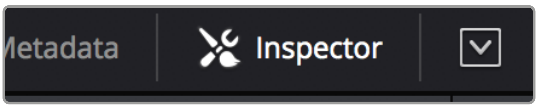
Modifiers displayed in the Modifiers panel

Customizing the Inspector

You can customize how the Inspector is presented in a variety of ways.

Inspector Height

A small arrow button at the far right of the UI toolbar lets you toggle the Inspector between full-height and half-height views, depending on how much room you need for editing parameters.

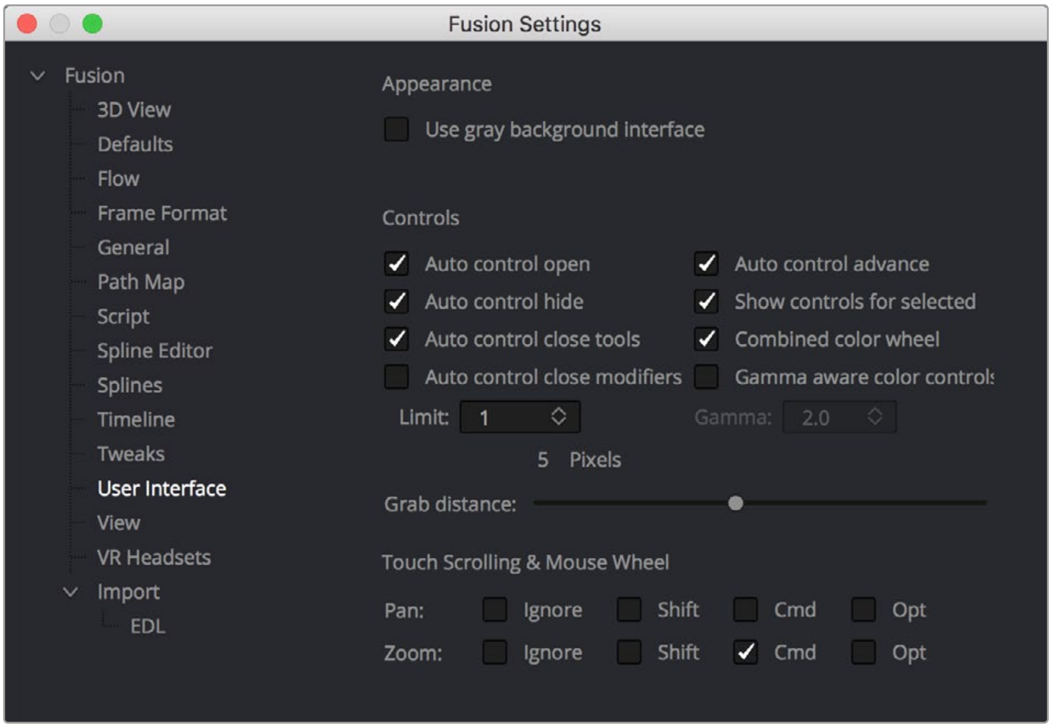


The Maximize button on the left side of the Inspector

In maximized height mode, the Inspector takes up up the entire right side of the UI, letting you see every control that a node has available, or creating enough room to see the parameters of two or three pinned nodes all at once. In half-height mode, the top of the Inspector is aligned with the tops of the viewers, expanding the horizontal space that's available for the Node Editor.

Inspector Display Preferences

By default, you see only selected nodes in the Inspector, and only the Active node is expanded to show its controls. You can change this behavior by choosing Fusion > Fusion Settings in the Fusion page or File > Preferences in Fusion Studio and opening the User Interface panel. In the User Interface, checkboxes manage the display of controls.



Control preferences in the User Interface category

- **Auto Control Open:** When enabled (the default), whichever node is active automatically opens its controls in the Inspector. When disabled, selecting an active node opens that node's Inspector header in the Inspector, but the parameters remain hidden unless you click the Inspector header.
- **Auto Control Hide:** When enabled (the default), only selected nodes are visible in the Inspector, and all deselected nodes are automatically removed from the Inspector to reduce clutter. When disabled, parameters from selected nodes remain in the Inspector, even when those nodes are deselected, so that the Inspector accumulates the parameters of every node you select over time.
- **Auto Control Close Tools:** When enabled (the default), only the parameters for the active node can be exposed. When disabled, you can open the parameters of multiple nodes in the Inspector if you want.
- **Auto Controls for Selected:** When enabled (the default), selecting multiple nodes opens multiple control headers for those nodes in the Inspector. When disabled, only the active node appears in the Inspector; multi-selected nodes highlighted in white do not appear.

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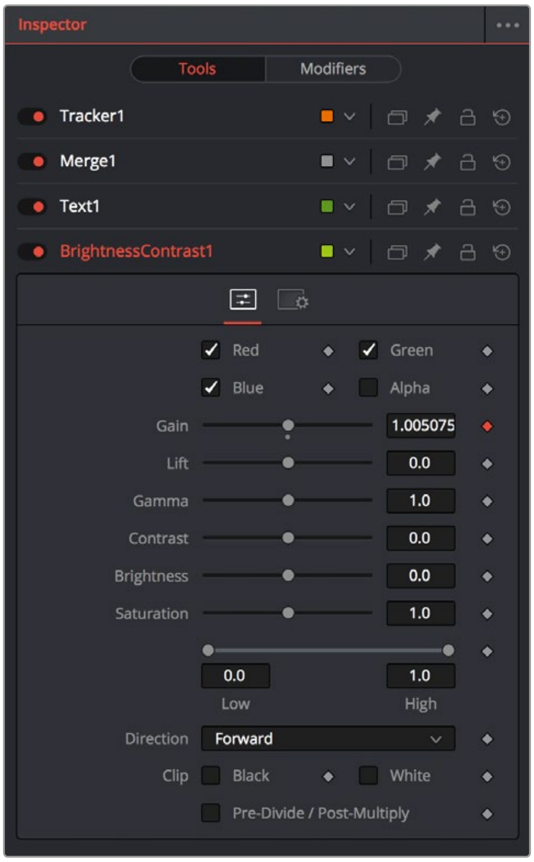
Opening Nodes in the Inspector

Before you can edit a node's parameters, you need to open it in the Inspector.

To display a node's controls in the Inspector:

Select one or more nodes from the Node Editor, Keyframes Editor, or Spline Editor.

When you select a single node so that it's highlighted orange in the Node Editor, all of its parameters appear in the Inspector. If you select multiple nodes at once, Inspector headers appear for each selected node (highlighted in white in the Node Editor), but the parameters for the active node (highlighted in orange) are exposed for editing.

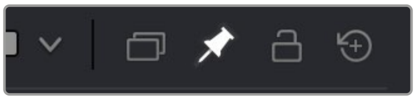


Opening multiple nodes in the Inspector

Only one node's parameters can be edited at a time, so clicking another node's Inspector header opens that node's parameters and closes the parameters of the previous node you were working on. This also makes the newly opened node the active node, highlighting it orange in the Inspector.

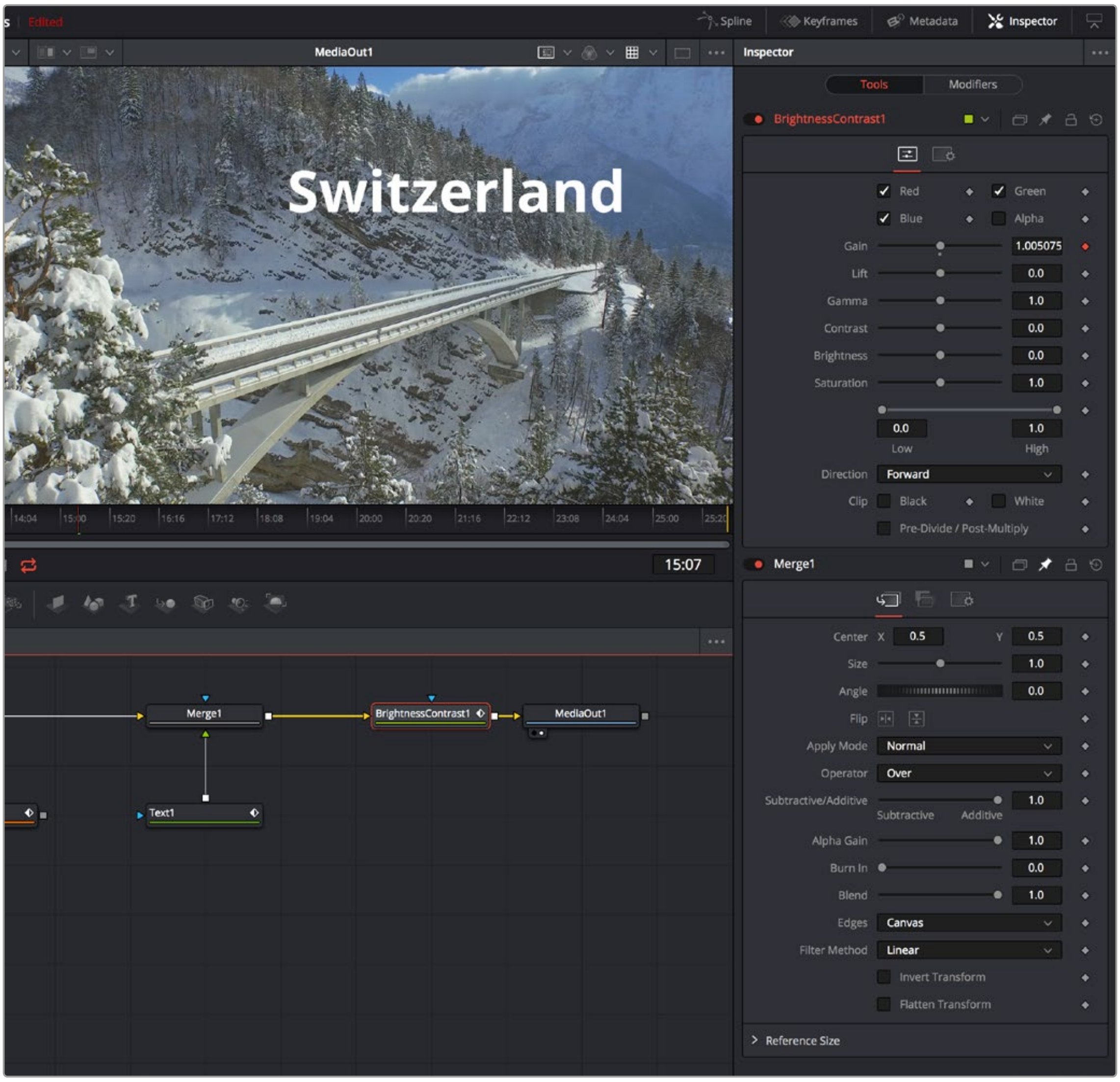
Pinning Multiple Nodes in the Inspector

For instances where you need to work quickly by editing the parameters of multiple nodes at the same time, you can use the Pin button in the Inspector header of nodes in the Inspector to keep those parameters exposed in the Inspector, regardless of whether that node is selected and active.



The Pin button of a node's Inspector header in the Inspector

While the Pin button is on, that node's parameters remain open in the Inspector. If you select another node in the Node Editor, that node's parameters appear beneath any pinned nodes.



A pinned node on the bottom, with a selected node at the top

You can have as many pinned nodes in the Inspector as you like, but the more you have, the more likely you'll need to scroll up or down in the Inspector to get to all the parameters you want to edit. To remove a pinned node from the Inspector, just turn off its Pin button in the Inspector header.

Hiding Inspector Controls

If you like, Inspector parameters for specific nodes can be hidden so they never appear, even when that node is selected. This can be useful for preventing accidental changes by you or other compositors who may be working on a composition in situations where you don't want to lock the node.

To Toggle the Inspector controls for a node or or off:

Right-click on the node in the Node Editor, or on the Inspector header, and choose Modes > Show Controls from the contextual menu.

Using the Inspector Header

When you select a node, it populates the Inspector with a title bar, or Inspector header, that displays that node's name as well as other controls that govern that node. A node's Inspector header itself has a variety of controls, but clicking (or double-clicking) on an Inspector header also exposes that node's parameters.

A node's Inspector header

When you select multiple nodes at once, you'll see multiple headers in the Inspector. By default, only the parameters for the active node (highlighted orange in the Node Editor) can be opened at any given time, although you can change this behavior in Fusion's Preferences.

Selecting and Viewing Nodes in the Inspector

Inspector headers are click targets for selecting nodes, opening and closing node parameters, and other things.

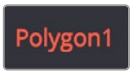
Methods of using headers:

- **To select a node using the Inspector header:** When multiple nodes are selected, you can make a node the active node by clicking its Inspector header in the Inspector. As the actively selected node, the Inspector header and the corresponding node in the Node Editor are highlighted orange, and its parameters are exposed.
- **To load a node into the viewer using the Inspector header:** You can view a node by dragging its header into one of the viewers.
- **To view a node's splines with the control header:** If you want to view the animated curves of a node in the Spline Editor, you can add them by dragging the Inspector header into the Spline Editor. All animated splines for the parameters of that node will automatically be displayed.

Using Header Controls

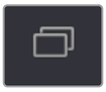
The controls found in each node's Inspector header makes it fast to do simple things.

To turn nodes off and on: Each Inspector header has a toggle switch to the left of its name, which can be used to enable or disable that node. Disabled nodes pass image data from the previous upstream node to the next downstream node without alteration.

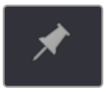


To change the Inspector header name: The name of the node corresponding to that Inspector header is displayed next. You can change the name by right-clicking the Inspector header to expose contextual menu commands similar to those found when you right-click a node in the Node Editor and choosing Rename. Alternatively, you can click an Inspector header and press F2 to edit its name. A Rename dialog appears, where you can enter a new name and click OK (or press Return).

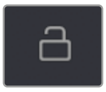
To color-code nodes: A color pop-up menu lets you color code with one of 16 colors. Choose Clear Color if you want to return that node to the default color.



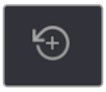
To version nodes: Turning on the Versions button displays a Version bar with six buttons. Versioning is described in the following section.



To pin Inspector controls: Clicking the Pin button "pins" that node's parameters in the Inspector so they remain in place, even if you deselect that node. You can have as many pinned nodes as you like in the Inspector, but the more you have, the more likely you'll be scrolling up and down the Inspector to navigate all the available parameters.



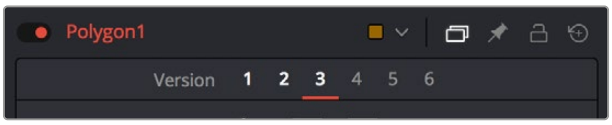
To lock nodes: Clicking the Lock button locks that node so no changes can be made to it.



To reset Inspector controls: The rightmost button in the Inspector header is a Reset button that resets the entire node to the default settings for that node.

Versioning Nodes

Each button is capable of containing separate parameter settings for that node, making it easy to save and compare up to six different versions of settings for each node. All versions are saved along with the node in the Node Editor for future use.

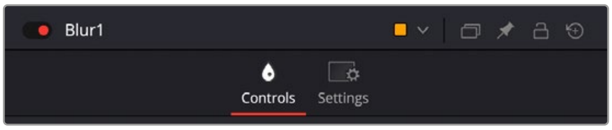


The Version bar, underneath an Inspector header, with versions enabled

An orange underline indicates the currently selected version, which is the version that's currently being used by your composition. To clear a version you don't want to use any more, right-click that version number and choose Clear from the contextual menu.

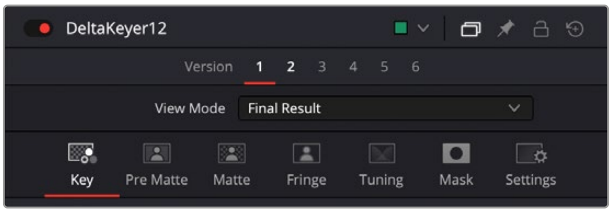
Parameter Tabs

Underneath the Inspector header is a series of panel tabs, displayed as thematic icons. Clicking one of these icons opens a separate tab of parameters, which are usually grouped by function. Simple nodes, such as the Blur node, consist of two tabs where the first contains all of the parameters relating to blurring the image, and the second is the Settings tab.



The parameter tabs of the Blur node

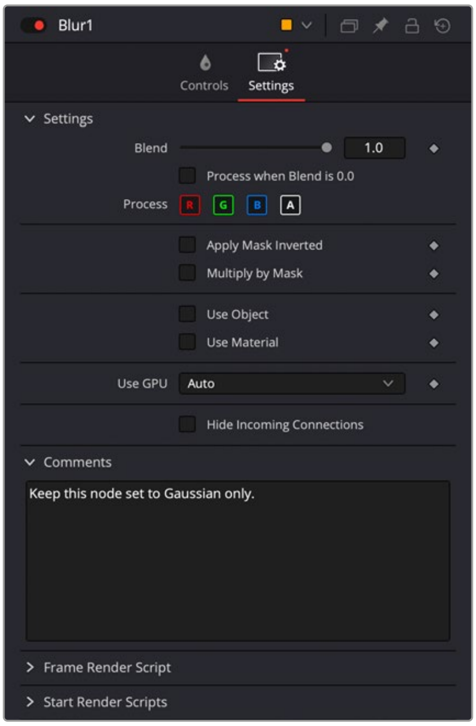
More complicated nodes have more tabs containing more groups of parameters. For example, the Delta Keyer has seven tabs: separating Key, Pre-Matte, Matte, Fringe, Tuning, and Mask parameters, along with the obligatory Settings tab. These tabs keep the Delta Keyer from being a giant scrolling list of settings and make it easy to keep track of which part of the keying process you're finessing as you work.



The parameter tabs of the Delta Keyer node

The Settings Tab

Every node that comes with Fusion has a Settings tab. This tab includes a set of standard controls that appear for nearly every node, although some nodes have special Settings tab controls that others lack.



The Settings tab in the Inspector

The following controls are common to most nodes, although some are node-specific. For example, Motion Blur settings have no purpose in a Color Space node.

Blend

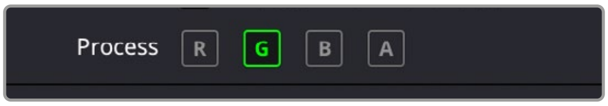
The Blend control is found in all nodes, except the Loader, MediaIn, and Generator nodes. It is used to blend between the node's unaltered image input and the node's final processed output. When the blend value is 0.0, the outgoing image is identical to the incoming image. Ordinarily, this will cause the node to skip processing entirely, copying the input straight to the output. The default for this node is 1.0, meaning the node will output the modified image 100%.

Process When Blend is 0.0

This checkbox forces the node to process even when the input value is zero and the image output is identical to the image input. This can be useful on certain nodes or third-party plugins that store values from one frame to the next. If this checkbox is disabled on nodes that operate in this manner, the node will skip being processed when the Blend is set to 0, producing incorrect results on subsequent frames.

Red/Green/Blue/Alpha Channel Checkboxes

Most nodes have a set of RGBA boxes in the Settings tab. These selectable boxes let you exclude any combination of these channels from being affected by that node.



The channel limiting boxes in the Settings panel of a Transform node set so that only the green channel is affected.

For example, if you wanted to use the Transform node to affect only the green channel of an image, you can turn off the Red, Blue, and Alpha checkboxes. As a result, the green channel is processed by this operation, and the red, blue, and alpha channels are copied straight from the node's input to the node's output, skipping that node's processing to remain unaffected.

Transforming only the green color channel of the image with a Transform effect

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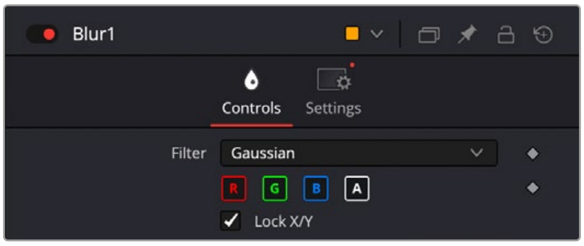
Skipping Channel Processing

Under the hood, most nodes actually process all channels first, but afterward copy the input image to the output for channels that have been unchecked. Modern workstations are so fast that this isn't usually noticeable, but there are some nodes where deselecting a channel actually causes that node to skip processing that channel entirely. Nodes that operate this way have a linked set of Red, Green, Blue, and Alpha boxes on another tab in the node.



Channel boxes on the Controls tab of the Blur node indicate that disabled channels won't be processed at all, to save rendering time..

In these cases, the Common Control channel boxes are instanced to the channel boxes found elsewhere in the node. Blur, Brightness/Contrast, Erode/Dilate, and Filter are examples of nodes that all have RGBY checkboxes in the main Controls tab of the Inspector, in addition to the Settings tab.

Apply Mask Inverted

When the Apply Mask Inverted checkbox is enabled, masks attached to the Effect Mask input of that node are inverted.

> **TIP:** The Apply Mask Inverted checkbox option operates only on effects masks, not on garbage masks.

Multiply By Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the Mask channel's values. This will cause all pixels of the image not included in the mask (i.e., those set to 0) to become black. This creates a premultiplied image.

Use Object/Use Material (For Masking)

Some 3D animation and rendering software can output to file formats that support auxiliary channels. Notably, the OpenEXR file format supports Object ID and Material ID channels, either of which can be used as a mask for an effect. This checkbox determines whether the channels will be used if they are available. The specific Material ID or Object ID affected is chosen using the next set of controls.

- **Sample Controls:** The Sample Controls are only displayed once the Use Object or Use Material checkbox is enabled. These controls select which ID is used to create a mask from the Object or Material channels saved in the image. You use the Sample button to grab IDs from the image in the viewer, the same way you use the Color Picker to select a color, by holding down the left mouse button on the Sample button, then dragging over to the viewer to the part of the image you want to select. The image or sequence must have been rendered from a 3D software package with those channels included.
- **Correct Edges:** The Correct Edges checkbox is only displayed once the Use Object or Use Material checkbox is enabled. When the Correct Edges checkbox is enabled, the Coverage and

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Background Color channels are used to separate and improve the effect around the edge of the object. When disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

Motion Blur

For nodes that are capable of introducing motion, such as Transform nodes, Warp nodes, and so on, the Motion Blur checkbox toggles the rendering of motion blur on or off for that node. When this checkbox is enabled, the node's predicted motion is used to produce the blur caused by a virtual camera shutter. When the control is disabled, no motion blur is created.

When Motion Blur is disabled, no additional controls are displayed. However, turning on Motion Blur reveals four additional sliders with which you can customize the look of the motion blur you're adding to that node.

- **Quality:** Determines the number of samples used to create the blur. The default quality setting of 2 will create two samples on either side of an object's actual motion. Larger values produce smoother results but will increase the render time.
- **Shutter Angle:** Controls the angle of the virtual shutter used to produce the Motion Blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects. The default value for this slider is 100.
- **Center Bias:** Modifies the position of the center of the motion blur. Adjusting the value allows for the creation of trail-type effects.
- **Sample Spread:** Adjusting Sample Spread modifies the weight given to each sample. This affects the brightness of the samples set with the Quality slider.

Scripting

Scripting fields are present on every node and contain one or more editable text fields that can be used to add scripts that process when that node is rendering. For more information on the contents of this tab, please consult the Scripting documentation.

Comments

A Comments field is found on every node and contains a single text field that is used to add comments and notes to that node. To enter text, simply click within the field to place a cursor, and begin typing.

When a note is added to a node, the comments icon appears in the Control Header and can be seen in a node's tooltip when the cursor is placed over the node in the Node Editor. The contents of the Comments tab can be animated over time, if required.

> Additional controls appear under this tab if the node is a Loader. For more information, *see Chapter 104, "Generator Nodes."* In the DaVinci Resolve Reference Manual or Chapter 44 in the Fusion Reference Manual.

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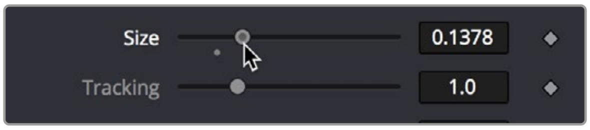
Inspector Controls Explained

Although a few nodes use fully customized interface elements that are unique to only that node, the vast majority of nodes use a mix of sliders, angle wheels, and checkboxes. This section explains how to use these controls.

Fusion Slider Controls

Slider Controls are used to select a single value from a range of values. You change the value by dragging the slider or entering a value into the edit box. This is fairly standard behavior for sliders. However, there is additional functionality that can increase your productivity when making changes with sliders.

Clicking on the gutter to the left or right of the handle will increase or decrease the value. Holding Command while clicking on the gutter will adjust the values in smaller increments. Holding Shift while clicking will adjust the value in larger increments.



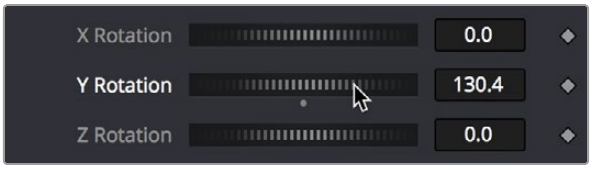
Hold Command while clicking in the gutter to move in smaller increments

While slider controls use a minimum and maximum value range, entering a value in the edit box outside that range will often expand the range of the slider to accommodate the new value. For example, it is possible to enter 500 in a Blur Size control, even though the Blur Size sliders default maximum value is 100. The slider will automatically adjust its maximum displayed value to allow entry of these larger values.

If the slider has been altered from its default value, a small circular indicator will appear below the gutter. Clicking on this circle will reset the slider to its default.

Thumbwheel

A Thumbwheel control is identical to a slider except it does not have a maximum or minimum value. To make an adjustment, you drag the center portion left or right or enter a value directly into the edit box. Thumbwheel controls are typically used on angle parameters, although they do have other uses as well.



Thumbwheel controls for X,Y, and Z rotation with arrows on either end for fine-tuning adjustments

Once the thumbwheel has been selected, you can use the Up and Down Arrows on your keyboard to further adjust the values. As with the slider control, the Command and Shift keys can be used to increase or decrease the change in value in smaller or larger increments.

If the thumbwheel has been altered from its default value, a small circular indicator will appear below above the thumbwheel. Clicking on this circle will reset the thumbwheel to its default.

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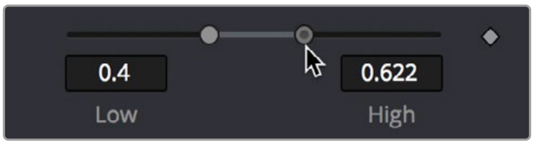
Range Controls

The Range controls are actually two separate controls, one for setting the Low Range value and one for the High Range value. To adjust the values, drag the handles on either end of the Range bar. To slide the high and low values of the range simultaneously, drag from the center of the Range bar. You can also expand or contract the range symmetrically by holding Command and dragging either end of the Range bar. You find Range controls on parameters that require a high and low threshold, like the Matte Control, Chroma Keyer, and Ultra Keyer nodes.

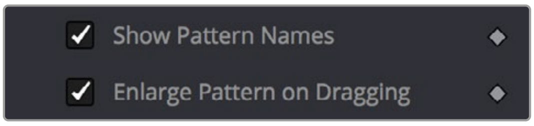


A Matte Threshold Range control

TIP: You can enter floating-point values in the Range controls by typing the values in using the Low and High numeric entry boxes.

Checkboxes

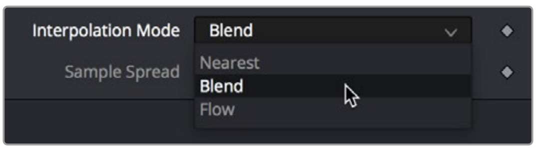
Checkboxes are controls that have either an On or Off value. Clicking on the checkbox control will toggle the state between selected and not selected. Checkboxes can be animated, with a value of 0 for Off and a value of 1.0 or greater for On.



Checkboxes used to select options for tracking

Drop-Down Menus

Drop-down menus are used to select one option from a menu. Once the menu is open, choosing one of the items will select that entry. When the menu is closed, the selection is displayed in the Inspector.



Drop-down menu in the TimeSpeed node

Drop-down menu selections can be animated, with a value of 0 representing the first item in the list, 1 representing the second, and so forth.

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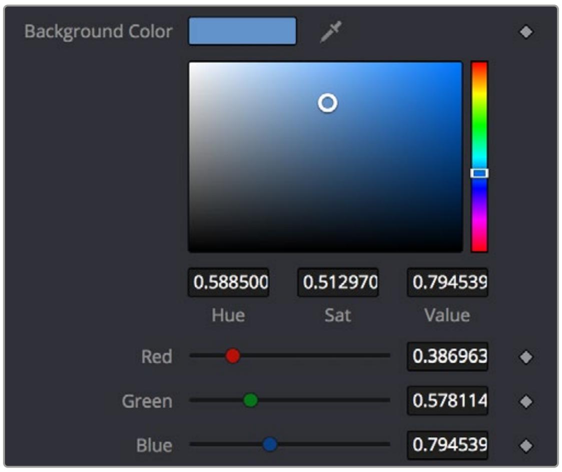
Button Arrays

Button arrays are groups of buttons that allow you to select from a range of options. They are almost identical in function to drop-down menu controls, except that in the case of a button array it is possible to see all of the available options at a glance. Often button arrays use icons to make the options more immediately comprehensible.

The Lens Type button array in the Defocus node

Color Chooser and Picker

The Color panel is displayed wherever a parameter requires a color as its value, such as the Fill or Outline color in the Text+ node. The selected color is shown in a swatch with an Eyedropper to its right, and below the swatch is the Color Chooser.



The Color panel with transparency preview

The Color panel is extremely flexible and has four different techniques for selecting and displaying colors.

TIP: Color can be represented by 0–1, 0.255, or 0–65000 by setting the range you want in the Preferences > General panel.

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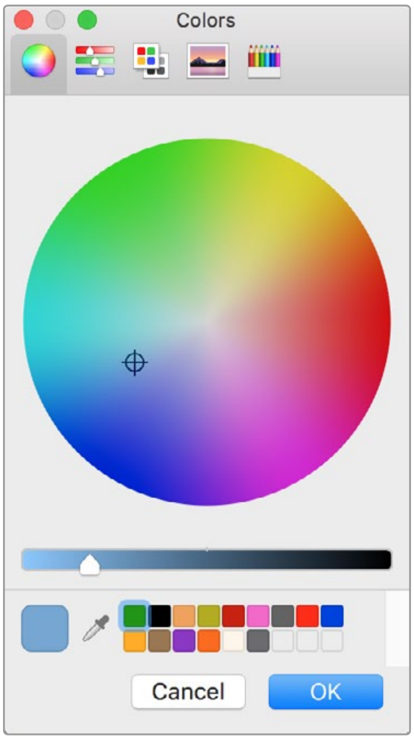
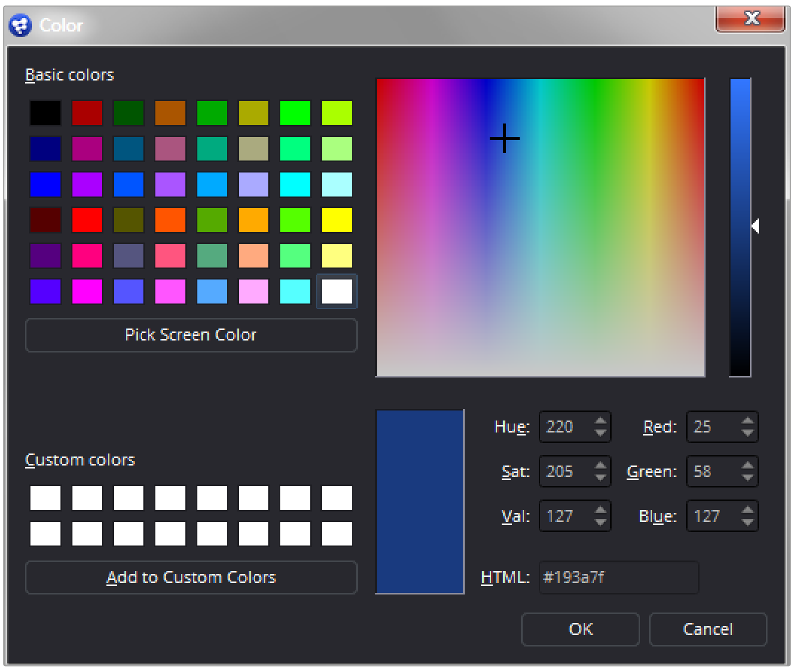
macOS and Windows Color Nodes

Clicking on the color swatch will display the operating system's standard Color Selection node.

Each operating system has a slightly different layout, but the general idea is the same. You can choose a color from the swatches provided—the color wheel on macOS, or the color palette on Windows. However you choose your color, you must click OK for the selection to be applied.



macOS Colors panel



Windows Color dialog

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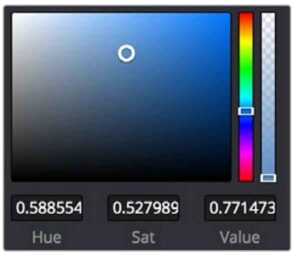
The Color Chooser

You also have access to the built-in color chooser, which includes sections for choosing grayscale values, as well as the currently chosen hue with different ranges of saturation and value. A hue bar and alpha bar (depending on the node) let you choose different values.



The color chooser in the Background node

Picking Colors from an Image

If you are trying to match the color from an image in the viewer, you can hold down the cursor over the Eyedropper, and then drag the pointer into the viewer. The pointer will change to an Eyedropper, and a pop-up swatch will appear above the cursor with the color you are hovering over and its values. When you are over the color you want, release the mouse button to set the color.

The Eyedropper with color swatch

The Color Picker normally selects from a single pixel in the image, but you can adjust the size of the selection by dragging into the viewer with the Eyedropper, and then holding Command and dragging out a rectangle for the sample size you want. The size change applies to all Color Pickers until the size is changed again.

Gradients

The Gradient Control bar is used to create a gradual blend between colors. The Gradient bar displays a preview of the colors used from start to end. By default, there are two triangular color stops: one on the left that determines the start color, and one on the right that determines the end color.

The default Gradient controls

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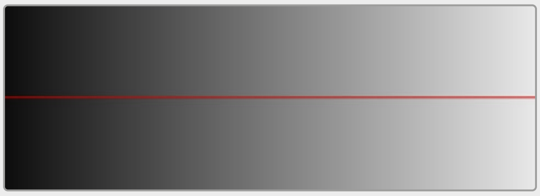
Gradient Type

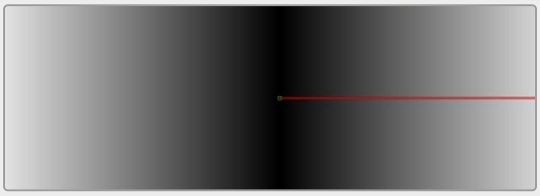
The Gradient Type button array is used to select the form used to draw the gradient.

Linear gradient: Linear draws the gradient along a straight line from the starting color stop to the ending color stop.



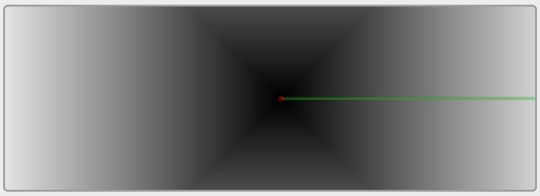
Linear gradient

Reflect gradient: Reflect draws the gradient by mirroring the linear gradient on either side of the starting point.



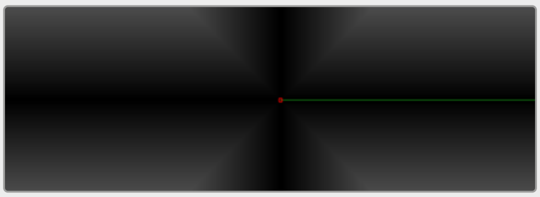
Reflect gradient

Square gradient: Square draws the gradient by using a square pattern when the starting point is at the center of the image.



Square gradient

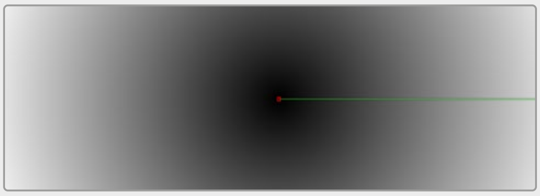
Cross gradient: Cross draws the gradient using a cross pattern when the starting point is at the center of the image.



Cross gradient

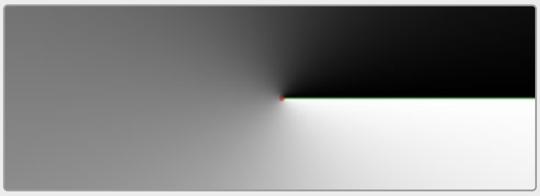
FUSION

Radial gradient: Radial draws the gradient in a circular pattern when the starting point is at the center of the image.



Radial gradient

Angle gradient: Angle draws the gradient in a counter-clockwise sweep when the starting point is at the center of the image.



Angle gradient

Start and End Position

The Start and End Position controls have a set of X and Y edit boxes that are useful for fine-tuning the start and end position of the gradient. The position settings are also represented by two crosshair onscreen controls in the viewer, which may be more practical for initial positioning.

Gradient Colors Bar

The Gradient Colors bar is used to select the blending colors for the gradient. The default two color stops set the start and end colors. You can change the colors used in the gradient by selecting the color stop, and then using the Eyedropper or color wheel to set the new color.

You can add, move, copy, and delete colors from the gradient using the Colors bar.

To add a color stop to the Gradient Colors bar:

1. Click anywhere along the bottom of the Gradient Colors bar.
2. Use the Eyedropper or color wheel to set the color for the color stop.

To move a color stop on the Colors bar:

- Drag a color stop left or right along the Gradient Color bar.

To copy a color stop on the Colors bar:

- Hold Command while you drag a color stop.

To delete a color stop from the Colors bar, do one of the following:

- Drag the color stop up past the Gradient Colors bar.
- Select the color stop, then click the red X button to delete it.

Interpolation Space

The Gradient Interpolation Method pop-up menu lets you select what color space is used to calculate the colors between color stops.

Offset

When you adjust the Offset control, the position of the gradient is moved relative to the start and end markers. This control is most useful when used in conjunction with the repeat and ping-pong modes described below.

Once/Repeat/Ping-Pong

These three buttons are used to set the behavior of the gradient when the Offset control scrolls the gradient past its start and end positions. The Once button is the default behavior, which keeps the color continuous for offset. Repeat loops around to the start color when the offset goes beyond the end color. Ping-pong repeats the color pattern in reverse.

1x1, 2x2, 3x3, 4x4, 5x5

These buttons control the amount of sub-pixel precision used when the edges of the gradient become visible in Repeat mode, or when the gradient is animated. Higher settings will take significantly longer to render but will be more precise.

Gradient Contextual Menu

Gradients have their own contextual menu that you can bring up by right-clicking on the Gradient bar. In the Gradient contextual menu are options for animating, publishing, and connecting one gradient to another. There is also a gradient-specific modifier that builds a custom gradient by sampling colors from the output of a node in the Node Editor.

Modifiers

Modifiers are expressions, calculations, trackers, paths, and other mathematical components that you attach to a parameter to extend its functionality. When a modifier is attached to a parameter, its controls will appear separately in the Inspector Modifiers tab.

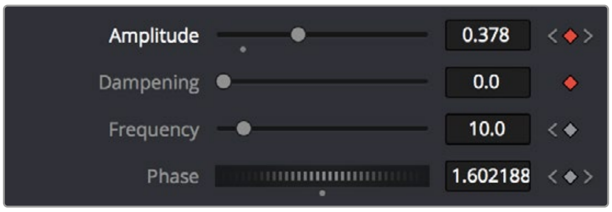
To attach a modifier:

1. Right-click over the parameter to which you want to attach a modifier.
2. Make a selection from the Modifier submenu in the contextual menu.

Animating Parameters in the Inspector

Fusion can keyframe most parameters in most nodes, in order to create animated effects such as animated transforms, rotoscoping with splines, dynamically altering warping behaviors, and so on; the list is endless. For convenience, a set of keyframing controls are available within the Inspector next to each keyframable parameter. These controls are:

- A gray Keyframe button to the right each keyframable parameter. Clicking this gray button creates a keyframe at the current position of the playhead, and turns the button orange.
- When you add a keyframe to a parameter, moving to a new frame and changing the parameter will automatically add a keyframe at the current position.
- Whenever the playhead is sitting right on top of a keyframe, this button turns orange. Clicking an orange Keyframe button deletes the keyframe at that frame and turns the button gray again.
- Small navigation arrows appear to the right and left if there are more keyframes in those directions. Clicking on navigation arrows to the right and left of keyframes jumps the playhead to those keyframes.



Orange Keyframe buttons in the Inspector show there's a keyframe at that frame

Once you've keyframed one or more parameters, If Show Modes/Options has been enabled, the node containing the parameters you keyframed displays a Keyframe badge, to show that node has been animated.

A keyframed node displays a Keyframe badge in the Node Editor

Once you've started keyframing node parameters, you can edit their timing in the Keyframes Editor and/or Spline Editor.

For more information about keyframing in Fusion, *see Chapter 71, "Animating in Fusion's Keyframes Editor,"* in the DaVinci Resolve Reference Manual or Chapter 9 in the Fusion Reference Manual.

Removing Animation From a Parameter

To remove all keyframes from a parameter:

1. Right-click over the name of the keyframed parameter in the Inspector.
2. Choose Remove "node name:parameter name" from the contextual menu.

> **TIP:** If you change the default spline type from Bézier, the contextual menu will display the name of the current spline type.

Attaching a Parameter to an Existing Animation Curve

Multiple parameters can be connected to the same animation curve. This can be an invaluable timesaver if you are identically animating different parameters in a node.

To connect a second parameter to the same animation curve:

1. Right-click on the second parameter you want to attach.
2. In the contextual menu, hover over the Connect To submenu.
3. In the Connect To submenu, choose the name of the animated parameter.

Connecting Parameters

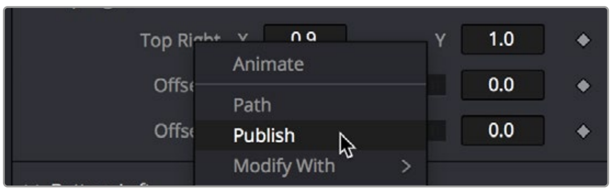
It is often useful to connect two parameters together even without an animation curve. There are two methods you can use.

Connecting Parameters by Publishing

If you want to tie two parameters together so adjusting one adjusts the other, you must connect them together using the Publish menu command on the first parameter and the Connect menu command on the second parameter.

To Publish and Connect parameters:

1. Right-click the name of the parameter you want to publish, and choose Publish from the contextual menu.
2. Right-click on the second parameter you want to attach, and choose the name of the parameter you just published from the Connect To submenu.



The Publish contextual menu

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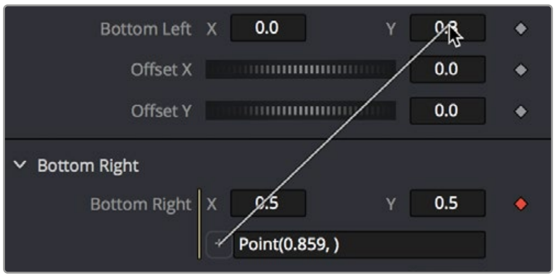
Connecting Parameters by Pick Whipping

You can also use simple expressions to link two parameters together. By using simple expressions via pick whipping, values can be connected and combined visually without the need to publish a value first. The pick whip is a temporary line drawn from one parameter to another in order to create a link between the two.

To link two parameters using a pick whip:

1. Double-click the field of a parameter you want to pick whip to another parameter, type =, and then press the Return key.
2. When Pick Whip controls appear underneath the parameter, drag a "whip" from the Add button to the target parameter.

   Now, adjusting the target parameter automatically adjusts the original parameter.



Pick whipping one parameter to another

TIP: Disabling the Auto Control Close node's General preference, and then selecting two nodes in the Node Editor will allow you to pick whip two parameters from different nodes.

The Expression field can further be used to add mathematical formulas to the value received from the target parameter.

For more information on Pick Whipping and Expressions, *see Chapter 74, "Using Modifiers, Expressions, and Custom Controls,"* in the DaVinci Resolve Reference Manual or Chapter 12 in the Fusion Reference Manual.

Contextual Menus

There are two types of contextual menus you can invoke within the Inspector.

Node Contextual Menus

To display the Node Context menu from the Inspector, right-click on the Inspector header. The node's contextual menu includes the same menu options that are accessed by right-clicking on a node in the Node Editor.

For more information *see Chapter 67, "Working in the Node Editor."* in the DaVinci Resolve Reference Manual or Chapter 5 in the Fusion Reference Manual for more information on these options.

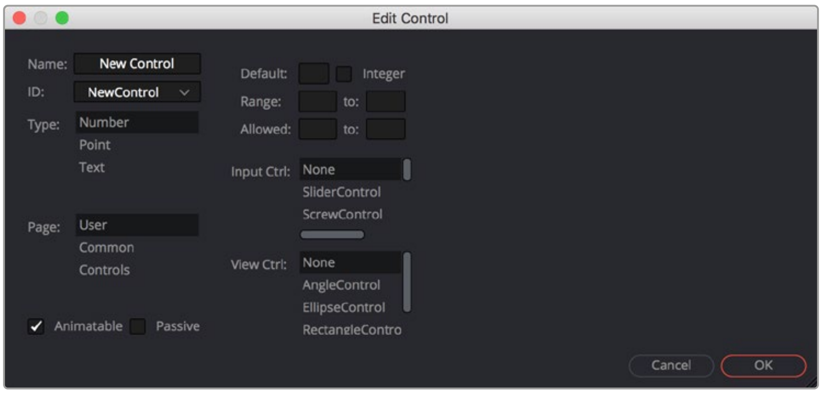
Parameter Contextual Menus

The contextual menu for individual parameters is accessed by right-clicking over the parameter's name, slider, thumbwheel, range control, button array, or other control type. For example, right-clicking on a slider will provide the slider's contextual menu, with options to animate the control or add additional modifiers. Many of these options were described in this chapter.

Customizing Node Parameters with User Controls

The user interface for each node in Fusion is designed to provide access to the parameters in a logical manner. Sometimes, though, you may want to add, hide, or change the controls. This is commonly done for simple expressions and macros, but it can be done for usability and aesthetic reasons for favorites and presets.

User custom controls can be added or edited via the Edit Control dialog. Right-click the name of a node in the Inspector (in the header bar) and choose Edit Control from the contextual menu. A new window will appear, titled Edit Control.



The Edit Control window

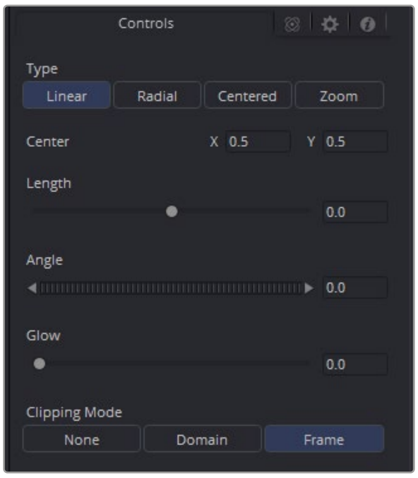
In the Input attributes, you can select an existing control or create a new one, name it, define the type, and assign it to a tab. In the Type attributes, you define the input controls, the defaults and ranges, and whether it has an onscreen preview control. The Input Ctrl attributes box contains settings specific to the selected node control, and the View Ctrl attributes box contains settings for the preview control, if any.

All changes made using UserControls are stored in the node instance itself, so they can be copy/pasted, saved to a setting, added to the Bins, or added to your favorites.

FUSION

An Example of Customizing Directional Blur

In the following example, let's suppose we wanted to create a more intuitive way of controlling a linear blur than using the Length and Angle sliders independently.



Default Directional Blur controls in the Inspector

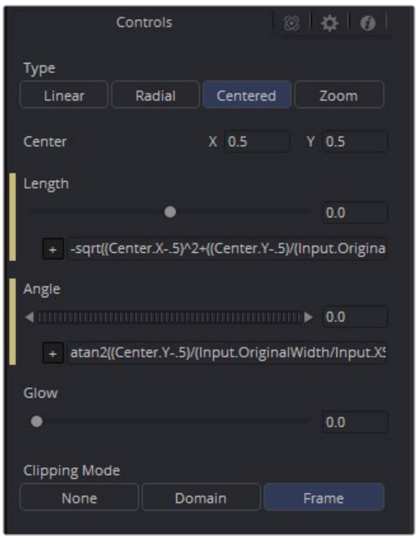
We could use the Center input control, along with its preview control, to set an angle and distance from directly within the viewer using expressions.

1. Right-click the label for the Length parameter, choose Expression from the contextual menu, and then paste the following expression into the Expression field that appears:

```
-sqrt(((Center.X-.5)*(Input.XScale))^2+((Center.Y-.5)*(Input.YScale)*(Input.
Height/Input. Width))^2)
```

2. Next, right-click the label for the Angle parameter, choose Expression from the contextual menu, and then paste the following expression into the Expression field that appears:

```
atan2(Center.Y-.5)/(Input.OriginalWidth/Input.X , .5-Center.X) * 180 / pi
```



Directional Blur controlled by the Center's position

This functions fine, but the controls are confusing. The Center control doesn't work as the center anymore, and it should be named "Blur Vector" instead. The controls for the Length and Angle aren't meant to be edited, so they should be hidden away, and we're only doing a linear blur, so we don't need the buttons for Radial or Zoom. We just need to choose between Linear and Centered.

Adding Another Control

For the first task, let's rename the Center. From the Add Control window, select Center from the ID list. A dialog will appear asking if you would like to Replace, Hide, or Change ID. We'll choose Replace. Now we are editing the Center input. We'll change the Name to Blur Vector, set the Type to Point, and the Page to Controls, which is the first tab where the controls are normally. Press OK, and our new input will appear on our node in the Node Editor. The ID of the control is still Center, so our SimpleExpressions did not change.

To hide the Length and Angle, we'll run the UserControls script again. This time when we select the Length and Angle IDs, we'll choose Hide in the dialog. Press OK for each.

Finally, to change the options available in the Type, we have two options. We can hide the buttons and use a checkbox instead, or we can change the MultiButton from four entries to two. Let's try both.

- To add the checkbox, run UserControls again, but this time instead of selecting an existing ID, we'll type Centered into the Name. This will set the name and the ID of our input to Centered. The Type is set to Number, and the Page is set to Controls. Now in the Type Attributes, set the Input Ctrl to be CheckboxControl. Press OK, and now we have our checkbox. To make the new control affect the Type, add a SimpleExpression to the Type:

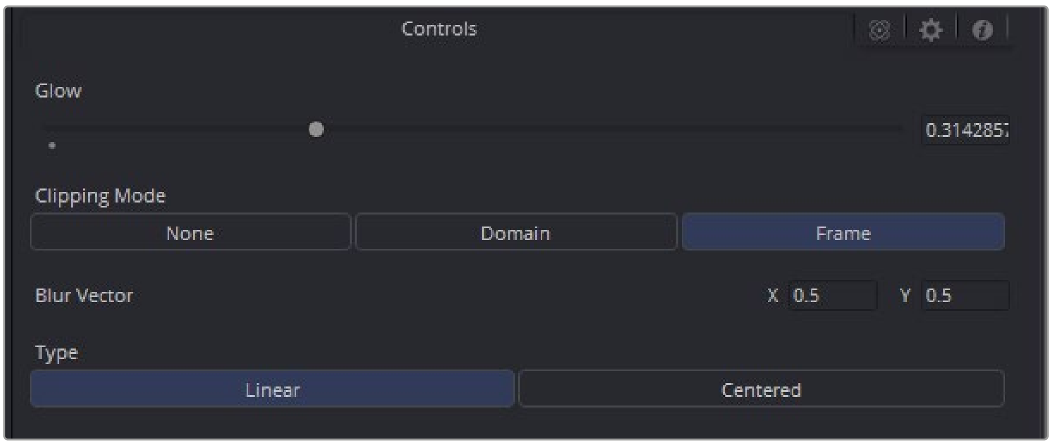
  ```
  iif(Centered==1, 2, 0).
  ```

  Once that's done, we can use the UserControls to hide the Type control.

- To make a new MultiButton, run the UserControl script, and add a new control ID, TypeNew. You can set the Name to be Type, as the Names do not need to be unique, just the IDs. Set the Type to Number, the Page to Controls, and the Input Ctrl to MultiButtonControl. In the Input Ctrl attributes, we can enter the names of our buttons. Let's do Linear and Centered. Type them in and hit Add for each. Press OK, and we have our new buttons with the unneeded options removed. To make this new control affect the original Type, add a SimpleExpression to the Type:

  ```
  iif(TypeNew==0, 0, 2).
  ```

  Once that's done, we can use the UserControls to hide the original Type control.



Directional Blurs with UserControls applied